# ML Research Project Guidelines

**Goal:** Students must clearly define the "What," "Why," and "How" of their machine learning project. What: Clearly state the problem being solved and the scope of the study. Why: Justify the relevance of the problem by identifying a specific and well-defined research gap supported by literature. How: Demonstrate how the proposed solution addresses the gap through a well-designed methodology and system architecture, followed by experiments, results, and critical discussion.

**1. Brainstorming and Problem Identification**

- **Action:** Groups must select a broad area of interest.
- **The background:** Students must briefly explain why this area matters in today's context.
- Students shouldn't just pick a dataset (e.g., Titanic); they should pick a problem type

**2. Identifying the Research Gap**

- **Literature Review:** Students must find relevant peer-reviewed published articles.
- **The Gap:** They must answer:
    - Why isn't this problem solved yet?
    - Is current SOTA (State of the Art) too computationally expensive?
    - Is there a lack of labeled data for this specific problem?
    - Do existing models fail under specific cases?

**3. Methodology & Architecture**

- Model Selection: Justification of the chosen architecture (e.g., CNN, LSTM, XGBoost).
- Hyperparameter Tuning: Documentation of the search space (GridSearch, Bayesian Opt).
- Loss Functions: Detailed explanation of the mathematical objective (MSE, Cross-Entropy, etc.).

**4. Experiments & Results**

- Metrics: Students must use appropriate metrics. (Accuracy isn't enough; they should use F1-Score, ROC-AUC, etc.).
- Visualization: Confusion matrices, Training vs. Validation loss curves, and Feature Importance plots.

**5. Discussion & Conclusion**

- **Interpretation:** What do the results mean in the context of the "Gap" identified in the mid-term?
- **Limitations:** Acknowledge what the study couldn't cover.
- **Future Recommendations:** Suggesting what the next researcher should do.

**Detailed Project Structure Table**

| Section | Description |
|---|---|
| Title & Abstract | A concise summary of the entire project. |
| Introduction | Background, Problem Statement, and Objectives. |
| Literature Review | Analysis of existing research to find the "Gap." |
| Methodology | The "blueprint" of how the research was conducted. |
| Results | The raw findings (data, observations). |
| Discussion | Connecting findings back to the original problem. |
| References | Consistent citation style as in the LaTex template. |

**Submission Requirements**

- The paper must be written using the LaTeX template provided here: https://icca.aiub.edu/authorsguideline.php#paper-submission
- Each group must also submit a "Group Contribution Log", clearly stating:
  - Each member's contribution
  - Sections/tasks completed by each member
- Submit a zipped file with proper name (Example: J G01 final project.zip) which should include the following:
  - J G01 Paper.pdf
  - J G01 code.ipynb
  - J G01 Dataset.pdf (should include the dataset link)
  - J G01 Contribution Log.pdf

**Submission Deadline**

- 2nd May 2026, 11:59 PM

**Late Submission Policy**

- A 10% mark deduction per day will be applied for late submissions.

**Viva Examination**

Individual viva sessions will be conducted in the final term week before the exam week.

**Evaluation Criteria**

- The project will be evaluated based on the provided rubric.
- Ensure your report aligns with all rubric components (problem, gap, methodology, results, etc.).

## Research Project Rubric

| Criteria | Exceptional | Proficient | Developing |
|---|---|---|---|
| **The ML Gap** | Identifies a specific technical bottleneck | Identifies a general problem but solution is a standard application. | No clear technical motivation; "just running a model." |
| **Literature Review** | Comprehensive use of high-quality sources; synthesizes ideas rather than just listing them. | Uses required number of sources; provides a good summary of existing work. | Relies on few or low-quality sources; summary is repetitive. |
| **Data Handling** | Advanced preprocessing, handling imbalance, or data augmentation used. | Basic cleaning and train/test split performed correctly. | Data used "as-is" without understanding its distribution. |
| **Model Rigor** | Comparison against a baseline; evidence of hyperparameter tuning. | Model works, but tuning is minimal or undocumented. | Model overfits or underfits significantly with no explanation. |
| **Evaluation** | Uses multiple relevant metrics; includes error analysis. | Uses standard metrics (Accuracy/Loss) only. | Metrics are misinterpreted or missing. |
| **Reproducibility** | Code is modular, commented, and includes a requirements.txt or Notebook. | Code runs but is messy or lacks documentation. | Code is broken or plagiarized. |
| **Writing & Format** | Flawless academic tone; perfect citations; follows all formatting guidelines. | Professional tone; very few citation or grammatical errors. | Tone is inconsistent; several errors in citations or formatting. |